

Bonding/Molecular Structure

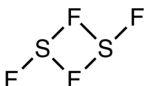
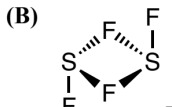
- Which gas-phase molecules are linear?
 - CO₂
 - ClO₂

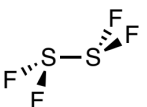
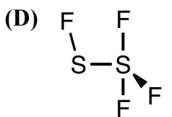
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Which best describes the formal charge on nitrogen in Lewis structures of the nitrite ion, NO₂⁻?

(A) 0
(B) +1
(C) -1
(D) 0 or -1, depending on the resonance structure
- Which species have a bond order of 3?
 - CO
 - CN⁻

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- How many σ and π bonds are there in a molecule of allene, H₂CCCH₂?

(A) 5 σ, 1 π (B) 5 σ, 2 π
(C) 6 σ, 1 π (D) 6 σ, 2 π
- Which best describes the arrangement of the atoms in space in S₂F₄?

(A)  (B) 

(C)  (D) 
- Which species have an octahedral geometry?
 - W(CO)₆
 - XeF₆

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Which factors contribute to inefficiencies in silicon-based photovoltaic devices in harvesting the energy of light from the sun?
 - Photons with energies less than the band gap of silicon are not absorbed by the device.
 - Energy is lost as heat when photons with energies greater than the band gap of silicon are absorbed by the device.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Which species has a square planar geometry?

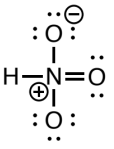
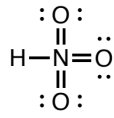
(A) BH₄⁻ (B) SF₄
(C) ZnCl₄²⁻ (D) AuCl₄⁻
- Which isomer of CNO⁻ is most stable?

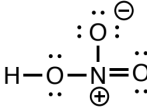
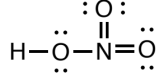
(A) Linear CNO⁻ (B) Linear NCO⁻
(C) Linear CON⁻ (D) Cyclic CNO⁻
- How many stereoisomers of octahedral Co(en)₂Cl₂⁺ are possible (en = ethylenediamine, H₂NCH₂CH₂NH₂)?

(A) 1 (B) 2 (C) 3 (D) 4
- Which statement most accurately describes the bonding in the acetyl cation, CH₃CO⁺?

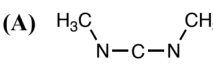
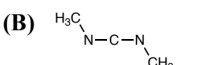
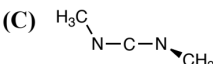
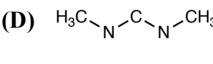
(A) The species has five σ bonds and two π bonds.
(B) The carbon-oxygen bond has a bond order of 2.
(C) The carbon bonded to oxygen has a bent geometry.
(D) The most significant resonance structure has a formal positive charge on carbon.
- Which statements about the gas-phase molecule NF are correct?
 - It has two unpaired electrons in its lowest-energy state.
 - It has a smaller ionization energy than atomic N.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Which is a valid Lewis structure for nitric acid, HNO₃?

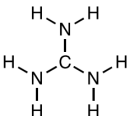
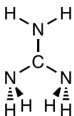
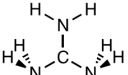
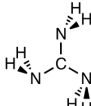
(A)  (B) 

(C)  (D) 

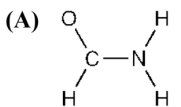
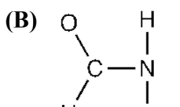
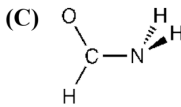
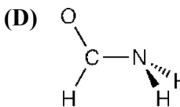
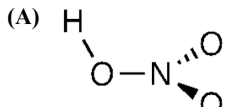
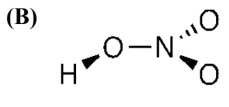
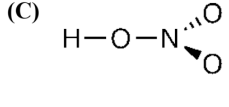
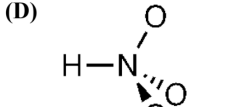
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14. Which statements about carbon-oxygen bond lengths are correct?
- The carbon-oxygen bond in carbon dioxide, CO_2 , is shorter than the carbon-oxygen bond in carbonyl sulfide, COS .
 - The carbon-oxygen bond in phosgene, COCl_2 , is shorter than the carbon-oxygen bond in urea, $\text{CO}(\text{NH}_2)_2$.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
15. Which statement best describes the relative Lewis acidity of BF_3 and BCl_3 ?
- (A) BF_3 is a weaker Lewis acid than BCl_3 because F is more electronegative than Cl.
(B) BF_3 is a weaker Lewis acid than BCl_3 because F is a better π donor than Cl.
(C) BF_3 is a stronger Lewis acid than BCl_3 because F is more electronegative than Cl.
(D) BF_3 is a stronger Lewis acid than BCl_3 because F is a better π donor than Cl.
16. Which comparisons of bond angles are correct?
- The bond angle in NH_3 is greater than the bond angle in NF_3 .
 - The bond angle in NH_3 is greater than the bond angle in PH_3 .
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
17. The gas-phase molecules B_2 and O_2 are both paramagnetic. Which statement about the relative energies of their molecular orbitals explains these observations?
- The σ_{2p} MO is higher in energy than the π_{2p} MO's.
 - The σ^*_{2p} MO is higher in energy than the π^*_{2p} MO's.
- (A) I for B_2 , II for O_2 (B) I for O_2 , II for B_2
(C) I for both B_2 and O_2 (D) II for both B_2 and O_2
18. How many stereoisomers are there of the octahedral coordination complex $\text{Co}(\text{NH}_2\text{CH}_2\text{CO}_2)_3$?
- (A) 1 (B) 2 (C) 3 (D) 4
19. Which species has the longest carbon-oxygen bond?
- (A) HCO_2^- (B) CO_3^{2-} (C) CO_2 (D) COS
20. Which molecule has a dipole moment of zero?
- (A) BCl_3 (B) NI_3 (C) AsH_3 (D) BrF_3
21. Which ions are linear?
- NO_2^+
 - ICl_2^-
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
22. Which is the best description of the structure of the substance whose empirical formula is NPF_{10} ?
- (A) $\text{F}_5\text{N}-\text{PF}_5$ (B) $\text{F}_3\text{N}-\text{PF}_7$
(C) $[\text{NF}_4][\text{PF}_6]$ (D) $[\text{PF}_4][\text{NF}_6]$
23. Which is the best representation of the three-dimensional arrangement of the atoms in dimethyl carbodiimide, $\text{CH}_3\text{NCNCH}_3$?
- (A)  (B) 
(C)  (D) 
24. Two substances with the formula $(\text{en})_2\text{CoCl}_3$ (en = ethylenediamine, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$) have different aqueous solubilities and different UV-visible spectra in aqueous solution. Which is the most likely explanation for this observation?
- (A) The substances are enantiomers.
(B) The substances are geometric isomers.
(C) One substance contains $[(\text{en})_2\text{CoCl}_2]\text{Cl}$ with six-coordinate cobalt and the other contains $(\text{en})_2\text{CoCl}_3$ molecules with seven-coordinate cobalt.
(D) One substance contains $[(\text{en})_2\text{CoCl}_2]\text{Cl}$ with six-coordinate cobalt and the other contains $[(\text{en})_2\text{Co}]\text{Cl}_3$ with four-coordinate cobalt.
25. Which diatomic molecule has the greatest bond dissociation enthalpy?
- (A) $\text{F}-\text{F}$ (B) $\text{Cl}-\text{Cl}$ (C) $\text{Br}-\text{Br}$ (D) $\text{I}-\text{I}$
26. Which molecule has the shortest carbon-nitrogen bond length?
- (A) HCNO (B) HNCO
(C) H_3CNO (D) H_3CNO_2

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27. Which statements about the structure of ClF_3 are correct?
- All $\text{F}-\text{Cl}-\text{F}$ angles are equal.
 - All $\text{Cl}-\text{F}$ bond lengths are equal.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
28. How many unpaired electrons are there in a ground-state NF molecule in the gas phase?
- (A) 0 (B) 1 (C) 2 (D) 4
29. The ionization energy of a ground-state nitrogen atom is 14.5 eV, while the ionization energy of a ground-state fluorine atom is 17.4 eV. What is the ionization energy of a ground-state NF molecule?
- (A) 13.1 eV (B) 15.1 eV
(C) 16.8 eV (D) 18.8 eV
30. Which is the best representation of the three-dimensional structure of the guanidinium ion, $\text{C}(\text{NH}_2)_3^+$?
- (A)  (B) 
- (C)  (D) 
31. Which complex ion absorbs visible light of the longest wavelength?
- (A) $\text{Co}(\text{NH}_3)_6^{3+}$ (B) $\text{Co}(\text{CN})_6^{3-}$
(C) CoF_6^{3-} (D) $\text{Co}(\text{NO}_2)_6^{3-}$
32. Which species has the largest $\text{O}-\text{N}-\text{O}$ bond angle?
- (A) NO_2 (B) N_2O_4 (C) NO_2^- (D) NO_3^-
33. Which best describes the isomer of the ion with the formula $[\text{CNO}]^-$ that is the most stable?
- (A) Linear, with a central C
(B) Linear, with a central N
(C) Linear, with a central O
(D) Cyclic
34. Which is the least important consideration in determining the strength of interaction between atomic orbitals on two atoms as they combine to form molecular orbitals?
- (A) How close the atoms are to one another
(B) How close in energy the orbitals are
(C) What the orbitals' relative orientation is
(D) Whether the orbitals have the same principal quantum number
35. Which statement about FNNF is NOT correct?
- (A) It exists as two distinct geometric isomers.
(B) It is planar.
(C) Its nitrogen-nitrogen bond is longer than that in N_2F_4 .
(D) It is more stable than its structural isomer with both fluorine atoms bonded to the same nitrogen.
36. A four-coordinate transition metal complex $\text{M}(\text{CO})_2(\text{Br})(\text{Cl})$ can be isolated as two geometric isomers. With which geometries at the metal center is this observation consistent?
- Square planar
 - Tetrahedral
- (A) I only (B) II only
(C) Either I or II (D) Neither I nor II
37. A sample of pure Si is doped with 1.0 part per million of As to form material **A**. Material **A** is then doped with 0.5 ppm of B to form material **B**. What is the correct order when the three materials are arranged in order of increasing electrical conductivity?
- (A) Pure Si < **A** < **B**
(B) **B** < **A** < Pure Si
(C) **B** < Pure Si < **A**
(D) Pure Si < **B** < **A**
38. Which species has the strongest bond?
- (A) HF (B) HCl (C) HBr (D) HI
39. Which species has the smallest bond angle?
- (A) NO_3^- (B) NO_2^- (C) FNO_2 (D) NF_3
40. Which ions are linear?
- N_3^-
 - $\text{N}_2\text{O}_2^{2-}$
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

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41. Which is the most stable arrangement of the atoms in space in formamide, HC(O)NH_2 ?
- (A)  (B) 
- (C)  (D) 
42. Molecular chlorine, Cl_2 , has a smaller ionization energy than atomic chlorine, Cl ($\text{IE} = 1108 \text{ kJ mol}^{-1}$ for Cl_2 and 1256 kJ mol^{-1} for Cl). Which is the best explanation for this difference?
- (A) Cl_2 has a covalent bond while Cl does not.
(B) Cl_2 has no unpaired electrons while Cl has one unpaired electron.
(C) The electron ionized from Cl_2 comes from an orbital that has antibonding character.
(D) The electron ionized from Cl_2 comes from an orbital that has d orbital character.
43. How many stereoisomers are possible for the complex ion $[\text{Co}(\text{NH}_3)_2(\text{CN})_2\text{Cl}_2]^-$?
- (A) 4 (B) 5 (C) 6 (D) 8
44. Which molecule has a dipole moment of zero?
- (A) CH_2Cl_2 (B) CHCl_3
(C) SO_2 (D) SO_3
45. The carbon-oxygen bond in CO has a higher bond dissociation enthalpy than a carbon-oxygen bond in CO_2 . Which is the best explanation for this difference?
- (A) CO has a bond order of 3 while each carbon-oxygen bond in CO_2 has a bond order of 2.
(B) CO is a polar molecule while CO_2 is a nonpolar molecule.
(C) CO contains one carbon-oxygen bond while CO_2 contains two carbon-oxygen bonds.
(D) CO has a lone pair on carbon while CO_2 does not.
46. What value is closest to the bond angle in the triiodide ion, I_3^- ?
- (A) 180° (B) 120° (C) 109.5° (D) 90°
47. Which is the best depiction of the three-dimensional arrangement of the atoms in nitric acid, HNO_3 ?
- (A)  (B) 
- (C)  (D) 
48. Which of these is NOT a correct statement about the ferrioxalate ion, $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$?
- (A) It is chiral.
(B) There are two different carbon-oxygen bond distances in the ion.
(C) There are two different iron-oxygen bond distances in the ion.
(D) There are six iron-oxygen bonds in the ion.
49. Which salt is diamagnetic?
- (A) $\text{K}_2[\text{NO}(\text{SO}_3)_2]$ (B) $\text{K}_4[\text{Fe}(\text{CN})_6]$
(C) $\text{Ce}_2(\text{SO}_4)_3$ (D) $\text{Hg}[\text{Co}(\text{SCN})_4]$
50. Which molecule has a nonzero dipole moment?
- (A) BF_3 (B) CF_4 (C) NF_3 (D) SF_6
51. What is the geometry of XeF_4 ?
- (A) Square planar (B) Tetrahedral
(C) See-saw (D) Square pyramidal
52. Which statements are correct about the cyanate ion, NCO^- ?
- I. Cyanate is more thermodynamically stable than its isomer fulminate (CNO^-).
II. The carbon-nitrogen bond is shorter than the carbon-oxygen bond in gas-phase cyanate ion.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
53. What is the coordination geometry of cobalt in the complex ion $[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_2\text{Br}_2]^+$?
- (A) Tetrahedral (B) Square planar
(C) Octahedral (D) Trigonal prismatic

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54. Sulfur tetrafluoride adopts a see-saw geometry with two axial F atoms with a F–S–F angle of about 180° and two equatorial F atoms at about 90° from the axial fluorines. Which statement most accurately describes the axial and equatorial S–F bonds?
- (A) The axial S–F bonds are longer because the two fluorines must share bonding to the same orbital on sulfur.
(B) The axial S–F bonds are longer because they experience greater repulsion from the other fluorine atoms in the molecule.
(C) The equatorial S–F bonds are longer because the equatorial F–S–F bond angle is the smallest in the molecule.
(D) The equatorial S–F bonds are longer because they experience greater repulsion from the lone pair on sulfur.
55. Which ion has the longest N–O bond?
- (A) NO_3^- (B) NO_2^- (C) NO_2^+ (D) NO^+
56. Which pair of species has the same shape?
- (A) CO_2 and SO_2 (B) CCl_4 and TiCl_4
(C) C_2H_6 and B_2H_6 (D) NO_3^- and PO_3^{3-}
57. How many distinct C–O bond lengths are present in the oxalate ion, $\text{C}_2\text{O}_4^{2-}$?
- (A) 1 (B) 2 (C) 3 (D) 4
58. Which statement about the molecular orbitals in a molecule is correct?
- (A) No molecular orbital may have a net overlap with any other molecular orbital.
(B) Each molecular orbital must have a different number of nodes than every other molecular orbital.
(C) The number of molecular orbitals is equal to half the number of atomic orbitals of the atoms that make up the molecule.
(D) The lowest-energy molecular orbitals are the most antibonding in character and the highest-energy molecular orbitals are the most bonding in character.
59. NF_3 has a bond angle of 102.5° , while PF_3 has a bond angle of 96.3° . What is the best explanation for the larger bond angle in NF_3 ?
- (A) The nitrogen 2s orbital participates more in bonding than does the phosphorus 3s orbital.
(B) Nitrogen is more electronegative than phosphorus.
(C) NF_3 has no unpaired electrons while PF_3 has two unpaired electrons.
(D) NF_3 is an ionic compound while PF_3 forms covalent bonds.
60. Allene has the structure $\text{H}_2\text{C}=\text{C}=\text{CH}_2$. What is the best description of the geometry of allene?
- | | Geometry at central carbon | Positions of hydrogen atoms |
|-----|----------------------------|-----------------------------|
| (A) | Linear | All in the same plane |
| (B) | Linear | In two perpendicular planes |
| (C) | Bent | All in the same plane |
| (D) | Bent | In two perpendicular planes |
61. What is the bond order of carbon monoxide, CO?
- (A) 1.5 (B) 2.0 (C) 2.5 (D) 3.0
62. Which statements about the Lewis structure of the fulminate ion, CNO^- , are correct?
- I. The nitrogen atom has a positive formal charge.
II. The nitrogen atom has a lone pair of electrons.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
63. In which molecule are there two distinct sets of sulfur-fluorine bond lengths?
- (A) SF_2 (B) SOF_2 (C) SF_4 (D) SF_6
64. Which species contains three sigma bonds and one pi bond?
- (A) PF_3 (B) NH_4^+ (C) C_2H_2 (D) CO_3^{2-}
65. How many stereoisomers of octahedral $\text{CoCl}_2(\text{NH}_3)_2(\text{CN})_2^-$ are possible?
- (A) 3 (B) 4 (C) 5 (D) 6

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66. What is the best description of the geometry of the nitrogen atoms in dimethylnitrosamine, $(\text{CH}_3)_2\text{NNO}$?

<u>N bonded to CH_3 groups</u>	<u>N bonded to O</u>
(A) Trigonal planar	Linear
(B) Trigonal planar	Bent
(C) Trigonal pyramidal	Linear
(D) Trigonal pyramidal	Bent

67. In the Lewis structure for the selenite ion, SeO_3^{2-} , how many lone pairs are around the central atom?

(A) 0 (B) 1 (C) 2 (D) 3

68. Removing an electron from molecular oxygen, O_2 , to form the dioxygenyl cation, O_2^+ , causes what changes in the bond length and in the number of unpaired electrons?

<u>Bond length</u>	<u>Number of unpaired electrons</u>
(A) Increase	Increase
(B) Increase	Decrease
(C) Decrease	Increase
(D) Decrease	Decrease

69. Which of the following molecules has a dipole moment of zero?

(A) HCN (B) CH_2Cl_2
(C) SO_2 (D) CO_2

70. Formamide has the structure HC(O)NH_2 . Which atoms in formamide have a trigonal planar geometry?

(A) C only (B) N only
(C) Both C and N (D) None of them

71. What is the coordination geometry of nickel in Ni(CO)_4 ?

(A) Tetrahedral (B) Square planar
(C) See-saw (D) T-shaped

72. Which of the following are ionic compounds?

I. NH_2NO_2 II. NH_4NO_3
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

73. When arranged in order of increasing bond strength, which order is correct?

(A) $\text{O-O} < \text{S-S} < \text{O=O} < \text{S=S}$
(B) $\text{O-O} < \text{S-S} < \text{S=S} < \text{O=O}$
(C) $\text{S-S} < \text{O-O} < \text{O=O} < \text{S=S}$
(D) $\text{S-S} < \text{O-O} < \text{S=S} < \text{O=O}$

74. Molecules with a permanent dipole moment include which of the following?

I. HCN II. O_3 III. XeF_2
(A) I only (B) I and II only
(C) II and III only (D) I, II, and III

75. How many stable resonance forms can be written for the oxalate ion, $\text{C}_2\text{O}_4^{2-}$?

(A) two (B) three (C) four (D) five

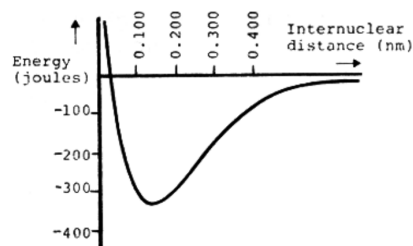
76. What is the geometry of ICl_4^- according to VSEPR theory?

(A) see-saw (B) square planar
(C) tetrahedral (D) T-shaped

77. According to the molecular orbital theory, what is the bond order of the He-H bond?

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) 1 (D) $1\frac{1}{2}$

78. As two atoms approach one another in space, the potential energy decreases then increases as shown in the diagram below.



To what are these two changes attributed?
[NOTE: p^+ = proton, e^- = electron]

<u>decrease</u> (from 0.40 to 0.15 nm)	<u>increase</u> (from 0.15 to 0 nm)
(A) increased $p^+ e^-$ attraction	increased $p^+ p^+$ repulsion
(B) increased $p^+ e^-$ attraction	increased $e^- e^-$ repulsion
(C) decreased $p^+ p^+$ repulsion	increased $e^- e^-$ repulsion
(D) decreased $p^+ e^-$ repulsion	increased $p^+ p^+$ repulsion

79. The infrared frequency of the CX vibration for CH_3X depends on which of the following?

I. mass of X
II. strength of the CX bond
III. type of CX vibration (stretch or bend)
(A) I. only (B) II. only
(C) II. and III. only (D) I., II. and III.

Bonding/Molecular Structure

80. How many sigma bonds are present in P_4 ?
 (A) 3 (B) 4 (C) 5 (D) 6
81. Using VSEPR theory, the molecular structure of IF_5 is best described as
 (A) octahedral (B) square pyramidal
 (C) trigonal bipyramidal (D) trigonal pyramidal
82. What is the formal charge on the oxygen atom in $:C:::O:$?
 (A) +1 (B) 0 (C) -1 (D) -2
83. Which is a valid Lewis structure for the nitrate ion, NO_3^- ?
 (A) $\left[\begin{array}{c} \ddot{O} = N = \ddot{O} \\ | \\ \ddot{O} \end{array} \right]^-$ (B) $\left[\begin{array}{c} \ddot{O} = N - \ddot{O} \\ | \\ \ddot{O} \end{array} \right]^-$
 (C) $\left[\begin{array}{c} \ddot{O} - N - \ddot{O} \\ | \\ \ddot{O} \end{array} \right]^-$ (D) $\left[\begin{array}{c} \ddot{O} = N = \ddot{O} \\ || \\ \ddot{O} \end{array} \right]^-$
84. Based on molecular orbital theory, which species has the highest bond order?
 (A) O_2^+ (B) O_2 (C) O_2^- (D) O_2^{2-}
85. How many carbon atoms can be described as sp^2 hybridized in this molecule?

$$\begin{array}{c}
 \text{O} \\
 || \\
 \text{H}_3\text{C}-\text{C}=\text{C}-\text{C}-\text{OH} \\
 | \quad | \\
 \text{H} \quad \text{H}
 \end{array}$$
 (A) 0 (B) 1 (C) 2 (D) 3
86. Which combination represents an n-type semiconductor?
 (A) Si doped with Ge (B) Si doped with As
 (C) Si doped with Ga (D) As doped with Ga
87. Which ionic compound has the smallest lattice energy?
 (A) LiI (B) NaF (C) $MgCl_2$ (D) MgO
88. What is the geometry of BrF_3 ?
 (A) seesaw (B) T-shaped
 (C) trigonal planar (D) trigonal pyramidal
89. Three monosulfur fluorides are known: SF_2 , SF_4 and SF_6 . Of these, polar species include
 (A) SF_2 only. (B) SF_4 only.
 (C) SF_2 and SF_4 only. (D) SF_2 , SF_4 and SF_6 .
90. Which reaction forms a product with a trigonal planar geometry?
 (A) $N_2 + 3H_2 \rightarrow$ (B) $2CO + O_2 \rightarrow$
 (C) $PCl_3 + Cl_2 \rightarrow$ (D) $2SO_2 + O_2 \rightarrow$
91. What is the best description of the hybridization of each of the carbon atoms (from left to right) in the compound $NCCH_2CO_2H$?
 (A) sp, sp^3, sp^2 (B) sp, sp^2, sp^3
 (C) sp^2, sp^3, sp^2 (D) sp^2, sp^3, sp^3
92. How many isomers exist for the square planar species $Pt(H_2O)(NH_3)ClBr$?
 (A) 1 (B) 2 (C) 3 (D) 4
93. When the molecules N_2 , N_2O , and N_2O_4 are arranged in order of decreasing N – N bond length, which order is correct?
 (A) N_2O_4, N_2O, N_2 (B) N_2, N_2O, N_2O_4
 (C) N_2O, N_2, N_2O_4 (D) N_2, N_2O_4, N_2O
94. From a consideration of the Lewis structure,
 $\left[\begin{array}{c} \ddot{N} = C = \ddot{O} \\ | \\ \ddot{O} \end{array} \right]^-$
 what are the formal charges?
 (A) $N = -1, C = 0, O = 0$ (B) $N = 0, C = 0, O = -1$
 (C) $N = -1, C = +1, O = -1$ (D) $N = -1, C = -1, O = +1$
95. What is the geometry of the IBr_2^- ion?
 (A) linear
 (B) bent with a bond angle of about 90°
 (C) bent with a bond angle of about 109°
 (D) bent with a bond angle of about 120°
96. How many isomers exist for the $[Co(NH_3)_4Cl_2]^+$ and $[Co(en)_2Cl_2]^+$ ions, respectively? [$en = H_2NC_2H_4NH_2$]
 (A) 2 and 2 (B) 2 and 3
 (C) 3 and 2 (D) 3 and 3

Bonding/Molecular Structure

97. Which statement best describes the structure of the allene molecule, $\text{H}_2\text{C}=\text{C}=\text{CH}_2$?

- (A) The C atoms form an angle of 120° and the H atoms lie in the same plane as the C atoms.
- (B) The C atoms form an angle of 120° and the H atoms lie in a plane perpendicular to that of the C atoms.
- (C) The C atoms form an angle of 180° and the four H atoms lie in the same plane.
- (D) The C atoms form an angle of 180° and the two CH_2 groups are perpendicular to one another.

98. Which molecule contains the smallest F-S-F angle?

- (A) SF_2 (B) SOF_2 (C) SO_2F_2 (D) SF_6

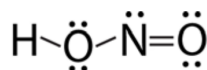
99. Which species has the longest N-O bond?

- (A) NO (B) NO^+ (C) NO_2 (D) NO_2^+

100. How many pi bonds and how many lone pairs are in the Lewis structure of hydrazine, N_2H_4 ?

- (A) 2 pi bonds, 0 lone pairs
- (B) 1 pi bond, 0 lone pairs
- (C) 1 pi bond, 1 lone pair
- (D) 0 pi bonds, 2 lone pairs

101. In the Lewis structure of nitrous acid:



What is the formal charge on nitrogen?

- (A) -1 (B) 0 (C) +1 (D) +3

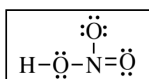
102. How many isomers of octahedral $\text{Co}(\text{NH}_3)_3\text{Cl}_3$ are there?

- (A) 2 (B) 3 (C) 4 (D) 5

103. The bond angle in H_2O is approximately 105° while the bond angle in H_2S is approximately 90° . Which explanation best accounts for this difference?

- (A) H-S bonds are longer than H-O bonds.
- (B) H-S bonds are less polar than H-O bonds.
- (C) S has d orbitals available for bonding, O does not.
- (D) O uses sp^3 hybrid orbitals for bonding, S uses its 3p orbitals.

104. What is the formal charge on the nitrogen atom in HNO_3 ?



- (A) 0 (B) +1 (C) +3 (D) +5

105. Which substance has the greatest lattice energy?

- (A) CaO (B) KCl (C) MgI_2 (D) BaS

106. Which species has a trigonal planar geometry?

- (A) ClF_3 (B) NCl_3 (C) CO_3^{2-} (D) I_3^-

107. Which of the molecules listed have an sp^3 hybridized central atom?

- | |
|---------------------|
| I. PCl_3 |
| II. COCl_2 |
| III. SF_4 |

- (A) I only (B) I and III only
- (C) II and III only (D) I, II and III

108. How many isomers exist for a square planar platinum compound which has four different groups attached to the platinum atom?

- (A) one (B) two (C) three (D) four

109. Molecules with non-zero dipole moments include which of those listed?

- | |
|--|
| I. $\text{H}_2\text{C}=\text{CHCl}$ |
| II. cis - $\text{ClHC}=\text{CHCl}$ |
| III. trans - $\text{ClHC}=\text{CHCl}$ |

- (A) I only (B) III only
- (C) I and II only (D) I, II and III

110. Which species is diamagnetic?

- (A) NO (B) N_2^+ (C) O_2 (D) O_2^{2-}

111. What is the I-I-I bond angle in the I_3^- ion?

- (A) 180° (B) 120° (C) 90°
- (D) more than 90° but less than 120°

112. Which species has the shortest nitrogen-oxygen bond?

- (A) NO^+ (B) NO_2^+ (C) NO_2^- (D) NO_3^-

113. Which substance will form hydrogen bonds to water molecules but will not form hydrogen bonds with its own molecules?

- (A) HF (B) $\text{C}_2\text{H}_5\text{OH}$
- (C) CH_3NH_2 (D) CH_3OCH_3

Bonding/Molecular Structure

114. In the gas phase PCl_5 exists as individual molecules but in the solid it takes on the ionic structure $\text{PCl}_4^+\text{PCl}_6^-$. What are the geometries of these three species

PCl_5	PCl_4^+	PCl_6^-
(A) trigonal bipyramidal	see-saw	octahedral
(B) trigonal bipyramidal	tetrahedral	octahedral
(C) trigonal bipyramidal	square planar	distorted octahedral
(D) square pyramidal	see-saw	square planar

115. Which species has the same electron distribution around the central atom as SiF_4 ?

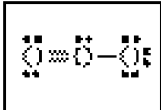
(A) SF_4 (B) XeF_4 (C) ClF_4^+ (D) BF_4^-

116. Which is/are polar species?

I. SF_2
II. SF_4
III. SF_6

(A) I only (B) III only
(C) I and II only (D) II and III only

117. According to the Lewis dot structure for ozone, what is the formal charge on the central oxygen atom?



(A) -2 (B) -1 (C) 0 (D) +1

118. When the species are arranged in order of increasing length of the carbon-oxygen bond, which order is correct?

(A) $\text{Na}_2\text{CO}_3 < \text{HCO}_2\text{Na} < \text{CH}_3\text{ONa}$
(B) $\text{CH}_3\text{ONa} < \text{HCO}_2\text{Na} < \text{Na}_2\text{CO}_3$
(C) $\text{HCO}_2\text{Na} < \text{Na}_2\text{CO}_3 < \text{CH}_3\text{ONa}$
(D) $\text{Na}_2\text{CO}_3 < \text{CH}_3\text{ONa} < \text{HCO}_2\text{Na}$

119. Which ionic solid would require the most energy to form gaseous ions?

(A) NaF (B) Na_2O (C) MgO (D) MgF_2

120. When the species listed are arranged in order of increasing bond angle, which order is correct?

(A) $\text{H}_2\text{Se}, \text{H}_2\text{S}, \text{H}_2\text{O}$ (B) $\text{H}_2\text{S}, \text{H}_2\text{Se}, \text{H}_2\text{O}$
(C) $\text{H}_2\text{S}, \text{H}_2\text{O}, \text{H}_2\text{Se}$ (D) $\text{H}_2\text{O}, \text{H}_2\text{S}, \text{H}_2\text{Se}$

121. What is the shape of the TeF_5^- anion?

(A) see-saw (B) square pyramidal
(C) trigonal pyramidal (D) trigonal bipyramidal

122. How many sigma and pi bonds are in maleic acid, $\text{HO}_2\text{CCHCHCO}_2\text{H}$?

(A) 7 sigma, 2 pi (B) 8 sigma, 3 pi
(C) 9 sigma, 2 pi (D) 11 sigma, 3 pi

123. How many isomers exist for the octahedral compound, $\text{Pt}(\text{NH}_3)_2\text{Cl}_4$?

(A) 1 (B) 2 (C) 3 (D) 4

124. What is the formal charge on the sulfur atom in SO_2 ? (Assume a Lewis dot structure in which all atoms obey the octet rule.)

(A) +1 (B) +2 (C) -1 (D) -2

125. Oxygen gas is paramagnetic. This observation is best explained by

(A) resonance.
(B) the Lewis structure of O_2 .
(C) the molecular orbital description of O_2 .
(D) the hybridization of atomic orbitals in O_2 .

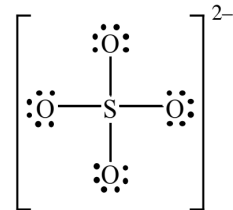
126. What is the geometry of the iodine atoms in the I_3^- ion?

(A) bent (B) linear
(C) T-shaped (D) triangular

127. Which species has a dipole moment other than zero?

(A) BrF_3 (B) CF_4 (C) SbF_5 (D) SF_6

128. In the Lewis structure what are the formal charges on the sulfur and oxygen atoms, respectively?

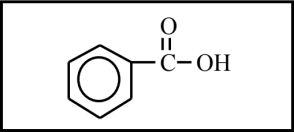


(A) 0, 0 (B) -2, 0 (C) +2, -1 (D) +6, -2

129. How many different isomers exist for the octahedral complex $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$?

(A) 1 (B) 2 (C) 3 (D) 4

Bonding/Molecular Structure

130. Which order is correct when the species are arranged in order of increasing average N-O bond length?
 (A) NO_3^- , NO_2^- , NO^+ (B) NO^+ , NO_3^- , NO_2^-
 (C) NO_2^- , NO_3^- , NO^+ (D) NO^+ , NO_2^- , NO_3^-
131. According to the Lewis dot structure shown, what are the formal charges of the O, C and N atoms, respectively, in the cyanate ion?
 $\text{:O}:::\text{C}::\ddot{\text{N}}:$
 (A) 0, 0, 0 (B) -1, 0, 0
 (C) -1, +1, -1 (D) +1, 0, -2
132. The hybridization of As in AsF_5 is best described as
 (A) sp^3 (B) sp^4 (C) dsp^3 (D) d^2sp^3
133. In which species do the atoms NOT lie in a single plane?
 (A) BF_3 (B) PF_3 (C) ClF_3 (D) XeF_4
134. The color of $\text{Co}(\text{H}_2\text{O})_6^{2+}$ is best attributed to electronic transitions
 (A) between different n levels in the metal.
 (B) between the metal's d orbitals.
 (C) from the Co^{2+} ion to water molecules.
 (D) during ionization.
135. When the carbon-oxygen bonds in the species; CH_3OH , CH_2O and CHO_2^- are arranged in order of increasing length, which is the correct order?
 (A) $\text{CH}_3\text{OH} < \text{CH}_2\text{O} < \text{CHO}_2^-$
 (B) $\text{CH}_2\text{O} < \text{CH}_3\text{OH} < \text{CHO}_2^-$
 (C) $\text{CHO}_2^- < \text{CH}_3\text{OH} < \text{CH}_2\text{O}$
 (D) $\text{CH}_2\text{O} < \text{CHO}_2^- < \text{CH}_3\text{OH}$
136. What is the total number of valence electrons in the peroxydisulfate, $\text{S}_2\text{O}_8^{2-}$, ion?
 (A) 58 (B) 60 (C) 62 (D) 64
137. For which species are both bonds of equal length?
 $\begin{matrix} 1. \text{ClO}_2^- \\ 2. \text{NO}_2^- \end{matrix}$
 (A) 1 only (B) 2 only
 (C) both 1 and 2 (D) neither 1 nor 2
138. Which compound has the highest melting point?
 (A) MgO (B) KCl (C) NaCl (D) CaO
139. Which molecular geometry is least likely to result from a trigonal bipyramidal electron geometry?
 (A) trigonal planar (B) see-saw
 (C) linear (D) t-shaped
140. Which diatomic species has the greatest bond strength?
 (A) NO (B) NO^+ (C) O_2 (D) O_2^-
141. During the complete combustion of methane, CH_4 , what change in hybridization does the carbon atom undergo?
 (A) sp^3 to sp (B) sp^3 to sp^2
 (C) sp^2 to sp (D) sp^2 to sp^3
142. Which oxides exist as individual molecules?
 1. Al_2O_3 2. SiO_2 3. P_4O_{10}
 (A) 2 only (B) 3 only
 (C) 1 and 3 only (D) 2 and 3 only
143. How many sigma and pi bonds are in this compound?

 (A) 9 sigma, 6 pi (B) 10 sigma, 6 pi
 (C) 10 sigma, 3 pi (D) 15 sigma, 4 pi
144. Which pair of ions has the same shape?
 (A) CO_3^{2-} and NO_3^- (B) CO_3^{2-} and SO_3^{2-}
 (C) NO_3^- and ClO_3^- (D) CO_3^{2-} and ClO_3^-
145. Which resonance form makes the greatest contribution to the structure of N_2O ?
 (A) $\ddot{\text{N}}::\text{N}::\text{O}::$ (B) $:\text{N}::\text{N}::\ddot{\text{O}}:$
 (C) $:\ddot{\text{N}}::\text{N}::\ddot{\text{O}}:$ (D) $:\text{N}::\text{N}::\ddot{\text{O}}:$
146. Which species has the strongest oxygen-oxygen bond according to molecular orbital theory?
 (A) O_2 (B) O_2^- (C) O_2^{2-} (D) O_2^+
147. How many atoms are covalently bonded to the chromium atom in $\text{Cr}(\text{NH}_3)_4\text{Cl}_3$?
 (A) 3 (B) 4 (C) 6 (D) 7

Bonding/Molecular Structure

148. When the carbon-oxygen bonds in H_3COH , H_2CO , and HCO_2^- are arranged in order of increasing length, what is the correct order?
- (A) H_3COH , H_2CO , HCO_2^-
 (B) HCO_2^- , H_3COH , H_2CO
 (C) H_2CO , HCO_2^- , H_3COH
 (D) H_3COH , HCO_2^- , H_2CO
149. What type of semiconductor results when highly purified silicon is doped with arsenic?
- (A) n -type (B) p -type
 (C) q -type (D) s -type
150. How many sigma and pi bonds are shown in this compound?
- $$\begin{array}{ccccccc} & \text{H} & \text{H} & \text{O} & & & \\ & | & | & || & & & \\ \text{H}-\text{N} & =\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{H} \\ & & | & & & & \\ & & \text{H} & & & & \end{array}$$
- (A) 8 sigma and 7 pi (B) 8 sigma and 3 pi
 (C) 11 sigma and 3 pi (D) 11 sigma and 4 pi
151. Which reaction involves a change in the electron-pair geometry for the underlined atom?
- (A) $\text{BF}_3 + \text{F}^- \rightarrow \text{BF}_4^-$ (B) $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$
 (C) $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$ (D) $\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{H}_3\text{O}^+$
152. Which compound has the largest lattice energy?
- (A) NaF (B) CsI (C) MgO (D) CaS
153. Which is the best description of a covalent bond?
- (A) Electrons are simultaneously attracted by more than one nucleus.
 (B) Filled orbitals of two or more atoms overlap one another.
 (C) Unoccupied orbitals of two or more atoms overlap one another.
 (D) Oppositely-charged ions attract one another.
154. What is the formal charge on the chlorine atom in the oxyacid HOClO_2 if it contains only single bonds?
- (A) -2 (B) -1 (C) +1 (D) +2
155. Which of these compounds is not adequately represented by a valence bond model?
- I. CO_2 II. SiO_2 III. SiO_2
- (A) I only (B) II only
 (C) I and III only (D) II and III only
156. Which species has the same shape as the NO_3^- ion?
- (A) SO_3 (B) SO_3^{2-}
 (C) ClF_3 (D) ClO_3^-
157. What is the formal charge on the central atom in N_2O ?
- $$\text{:N} \equiv \text{N} - \ddot{\text{O}}\text{:}$$
- (A) +1 (B) 0 (C) -1 (D) -2
158. How many bonding pairs and lone pairs surround the central atom in the I_3^- ion?
- | | Bonding Pairs | Lone Pairs |
|-----|---------------|------------|
| (A) | 2 | 2 |
| (B) | 2 | 3 |
| (C) | 3 | 2 |
| (D) | 4 | 3 |
159. The nitrogen atoms in NH_3 , NH_2^- , and NH_4^+ are all surrounded by eight electrons. When these three species are arranged in order of increasing H-N-H bond angle, what is the correct order?
- (A) NH_3 , NH_2^- , NH_4^+ (B) NH_4^+ , NH_2^- , NH_3
 (C) NH_3 , NH_4^+ , NH_2^- (D) NH_2^- , NH_3 , NH_4^+
160. What hybrid orbitals are employed by carbon atoms 1, 2, and 3, respectively, as labeled in the compound shown?
- $$\begin{array}{c} \text{:O:} \\ || \\ \text{H}_3\text{C}-\text{C}-\text{C} \equiv \text{N:} \\ \text{1} \quad \text{2} \quad \text{3} \end{array}$$
- (A) sp^3 , sp , sp (B) sp^2 , sp^2 , sp
 (C) sp^3 , sp^2 , sp (D) sp^3 , sp^2 , sp^2
161. In which pair, or pairs, is the stronger bond found in the first species?
1. O_2^- , O_2 2. N_2 , N_2^+ 3. NO^+ , NO^-
- (A) 1 only (B) 2 only
 (C) 1 and 3 only (D) 2 and 3 only
162. All of these species have the same number of valence electrons as NO_3^- *except*
- (A) CO_3^{2-} (B) HCO_3^- (C) NF_3 (D) SO_3
163. Which set contains no ionic species?
- (A) NH_4Cl , OF_2 , H_2S (B) CO_2 , Cl_2 , CCl_4
 (C) BF_3 , AlF_3 , TlF_3 (D) I_2 , CaO , CH_3Cl

Bonding/Molecular Structure

164. When these species are arranged in order of *increasing* bond energy, what is the correct sequence?

- (A) N_2 , O_2 , F_2 (B) F_2 , O_2 , N_2
(C) O_2 , F_2 , N_2 (D) O_2 , N_2 , F_2

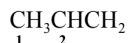
165. The geometry of the atoms in the species PCl_4^+ is best described as

- (A) tetrahedral (B) see-saw
(C) square (D) trigonal bipyramidal

166. Which are nonpolar molecules?

1. NCl_3 2. SO_3 3. PCl_5
(A) 1 only (B) 2 only
(C) 1 and 3 only (D) 2 and 3 only

167. What are the hybridizations of carbon 1 and carbon 2 in the hydrocarbon?



- (A) sp^3 , sp (B) sp^3 , sp^2
(C) sp^2 , sp^2 (D) sp , sp^2